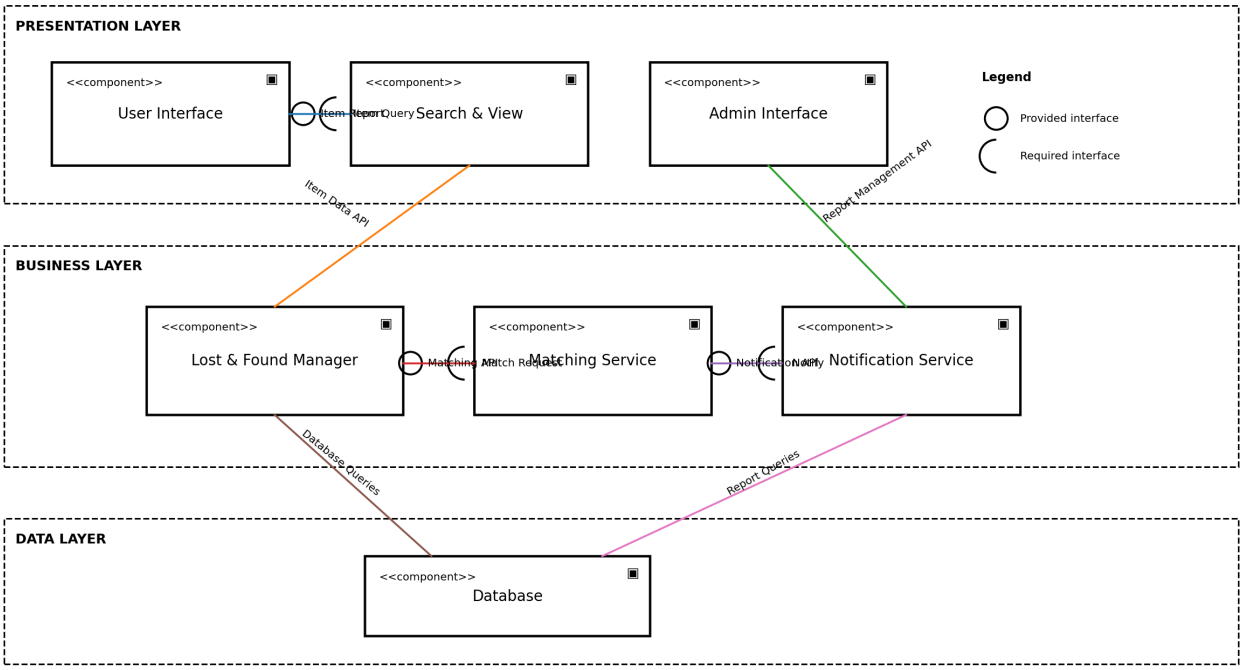
LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Community Lost & Found System | Layered Architecture

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Component Diagram



Community Lost & Found System — Layered Architecture

Architecture Selection & Justification

Architecture Selection

We chose Layered Architecture for the Community Lost & Found System. The architecture separates the presentation layer, business layer and data layer so that user interaction, application processing and data storage have distinct responsibilities.

Reason 1 – Separation of Concerns

The User Interface and Search/View components handle user interaction, while the Lost & Found Manager, Matching Service, Notification Service and Admin/Report Management components handle application functions. This keeps responsibilities organized and makes the system easier to understand and maintain.

Reason 2 – Maintainability

Core functions such as reporting items, matching lost and found records, sending notifications and managing reports are separated into components. A change to one responsibility can therefore be made with less impact on the other parts of the system.

Security Advantage

Database access is controlled through the business layer rather than allowing the user interface to directly access stored information. Validation and authorization can be applied at this controlled layer before reports or user data are stored or retrieved.

Performance Benefit

Separating search, matching and notification responsibilities allows each operation to be optimized independently. Database queries can be handled directly by the data layer while the business layer coordinates only the required operations.

Conclusion

Layered Architecture is appropriate for the Community Lost & Found System because it provides clear separation of responsibilities, straightforward maintenance and controlled access to stored data.